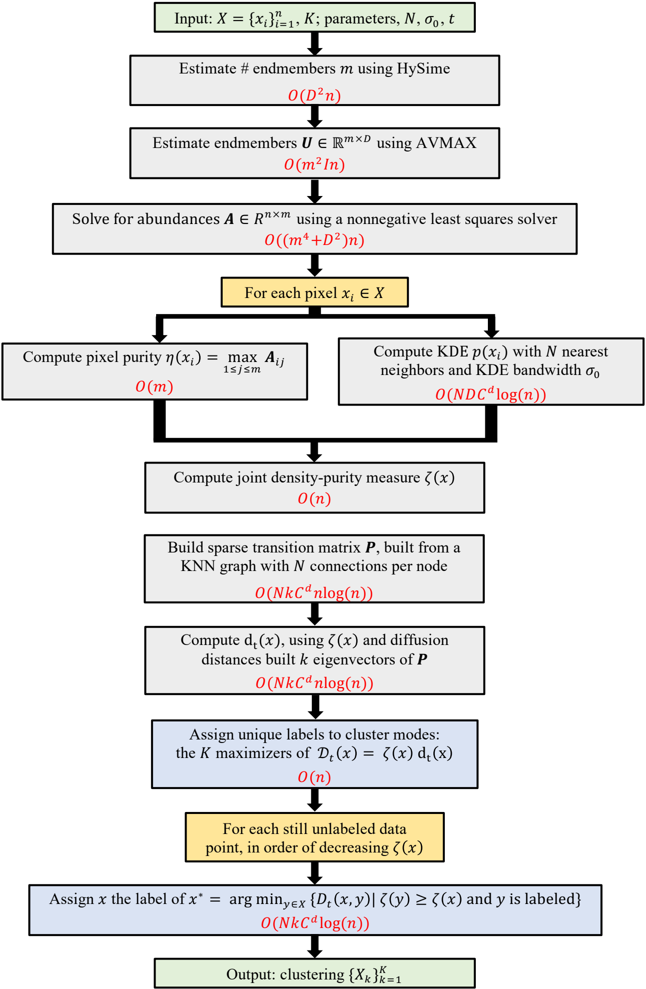
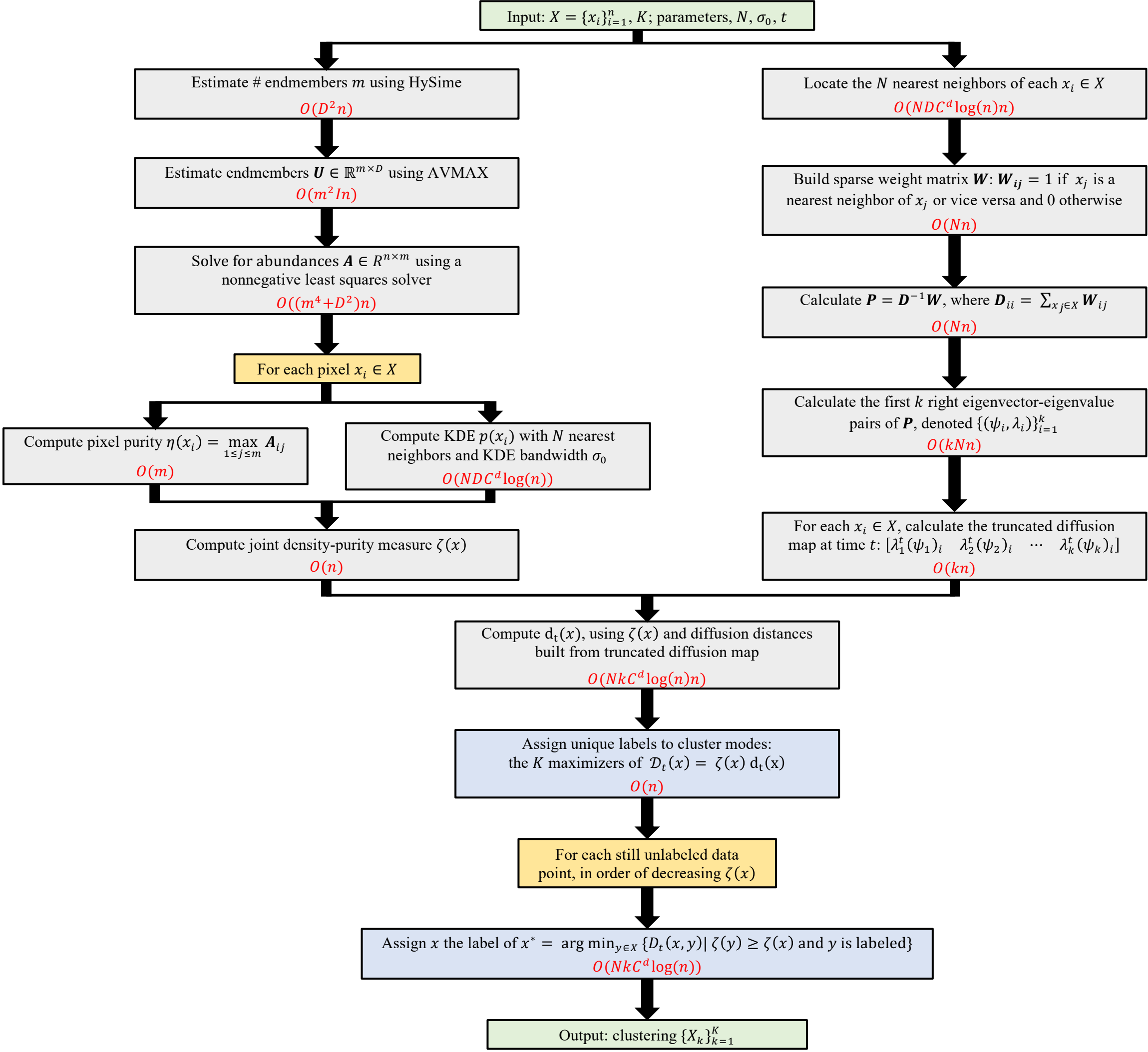


Input: $X = \{x_i\}_{i=1}^{n}$, K; parameters, N, σ_0, t
Estimate # endmembers m using HySime
$O(D^2 n)$
Estimate endmembers $\boldsymbol{U} \in \mathbb{R}^{m \times D}$ using AVMAX
$O(m^2 I n)$
Solve for abundances $\boldsymbol{A} \in R^{n \times m}$ using a nonnegative least squares solver
$O((m^4 + D^2)n)$
For each pixel $x_i \in X$
Compute pixel purity $\eta(x_i) = \max_{1 \le j \le m} \boldsymbol{A}_{ij}$
$O(m)$
Compute KDE $p(x_i)$ with N nearest neighbors and KDE bandwidth σ_0
$O(NDC^d \log(n))$
Compute joint density-purity measure $\zeta(x)$
$O(n)$
Build sparse transition matrix $\boldsymbol{P}$, built from a KNN graph with N connections per node
$O(NkC^d n\log(n))$
Compute $\mathrm{d}_t(x)$, using $\zeta(x)$ and diffusion distances built k eigenvectors of $\boldsymbol{P}$
$O(NkC^d n\log(n))$
Assign unique labels to cluster modes: the K maximizers of $\mathcal{D}_t(x) = \zeta(x)\,\mathrm{d}_t(\mathrm{x})$
$O(n)$
For each still unlabeled data point, in order of decreasing $\zeta(x)$
Assign x the label of $x^* = \arg\min_{y \in X} \{D_t(x,y) \mid \zeta(y) \ge \zeta(x)$ and y is labeled$\}$
$O(NkC^d \log(n))$
Output: clustering $\{X_k\}_{k=1}^{K}$


Input: $X = \{x_i\}_{i=1}^{n}$, K; parameters, N, σ_0, t
Estimate # endmembers m using HySime
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Estimate endmembers $\boldsymbol{U} \in \mathbb{R}^{m \times D}$ using AVMAX
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Solve for abundances $\boldsymbol{A} \in R^{n \times m}$ using a nonnegative least squares solver
$O((m^4 + D^2)n)$
For each pixel $x_i \in X$
Compute pixel purity $\eta(x_i) = \max_{1 \le j \le m} \boldsymbol{A}_{ij}$
$O(m)$
Compute KDE $p(x_i)$ with N nearest neighbors and KDE bandwidth σ_0
$O(NDC^d \log(n))$
Compute joint density-purity measure $\zeta(x)$
$O(n)$
Locate the N nearest neighbors of each $x_i \in X$
$O(NDC^d \log(n) n)$
Build sparse weight matrix $\boldsymbol{W}$: $\boldsymbol{W_{ij}} = 1$ if x_j is a nearest neighbor of x_j or vice versa and 0 otherwise
$O(Nn)$
Calculate $\boldsymbol{P} = \boldsymbol{D}^{-1}\boldsymbol{W}$, where $\boldsymbol{D}_{ii} = \sum_{x_j \in X} \boldsymbol{W}_{ij}$
$O(Nn)$
Calculate the first k right eigenvector-eigenvalue pairs of $\boldsymbol{P}$, denoted $\{(\psi_i, \lambda_i)\}_{i=1}^{k}$
$O(kNn)$
For each $x_i \in X$, calculate the truncated diffusion map at time t: $[\lambda_1^t(\psi_1)_i \quad \lambda_2^t(\psi_2)_i \quad \cdots \quad \lambda_k^t(\psi_k)_i]$
$O(kn)$
Compute $\mathrm{d}_t(x)$, using $\zeta(x)$ and diffusion distances built from truncated diffusion map
$O(NkC^d \log(n) n)$
Assign unique labels to cluster modes: the K maximizers of $\mathcal{D}_t(x) = \zeta(x)\, \mathrm{d}_t(\mathrm{x})$
$O(n)$
For each still unlabeled data point, in order of decreasing $\zeta(x)$
Assign x the label of $x^* = \arg\min_{y \in X} \{D_t(x, y) |\, \zeta(y) \ge \zeta(x)$ and y is labeled$\}$
$O(NkC^d \log(n))$
Output: clustering $\{X_k\}_{k=1}^{K}$